

Airway Obstruction from FB

Section 1: Case Summary

Scenario Title:	Airway obstruction
Keywords:	FB airway, respiratory arrest
Brief Description of Case:	This case involves a 60-year-old male patient who arrives VSA in PEA after collapsing while eating dinner with family. The collateral history included that he was suspected to be intoxicated. The patient arrives in PEA but is difficult to bag with EMS. The learner will have to work through the can't ventilate/can't oxygenate scenario once they identify that BVM is ineffective.

Goals and Objectives	
Educational Goal:	Management of respiratory arrest secondary to a food bolus.
Objectives: (Medical and CRM)	1. Recognize the need to place an advanced airway when BVM is not effective in cardiac arrest 2. Develop an approach to FB airway obstruction as a "can't intubate, can't ventilate" situation 3. Demonstrate situational awareness by delegating tasks, establishing leadership, and making use of a shared mental model

Learners, Setting and Personnel					
Target Learners:	☐ Junior Learners		☐ Senior Learners		☒ Staff
	☒ Physicians		☒ Nurses	☒ RTs	☐ Inter-professional
	☐ Other Learners:				
Location:	☐ Sim Lab		☒ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: 1				
	Confederates: 1				
	Sim Techs: 1				

Scenario Development	
Date of Development:	Jan 2021
Scenario Developer(s):	Rosamaria North, CCFP-EM and George McKay, CCFP-EM
Affiliations/Institutions(s):	Northern Ontario School of Medicine, Thunder Bay Regional Health Sciences Centre
Contact E-mail:	rbava@nosm.ca
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Jeffery Smith		Age: 60		Gender: M	Weight: 80 kg
Presenting complaint: VSA in PEA arrest					
Temp: 36.2	HR: 80	BP: --	RR: --	O ₂ Sat: 60%	FiO ₂ : 100 %
Cap glucose: --			GCS: 3 (E1 V1 M1)		
EMS handover (advanced care crew): The patient was apparently eating dinner with family. They witnessed the patient collapse and started CPR. The initial rhythm was PEA and the patient has remained in PEA. We've inserted an IV and given 3 doses of epinephrine. There is an OP airway in place but the patient is difficult to bag.					
Allergies: unknown					
Past Medical History: unknown			Current Medications: unknown		

Section 2B: Extra Patient Information

A. Further History	
This patient was eating dinner and had been intoxicated according to the family members. He was talking, then appeared to collapse according to witnesses. Family started CPR immediately.	
B. Physical Exam	
Cardio: No pulses	Neuro: GCS 3
Resp: Bagging with BVM and OP in place, there is poor chest rise	Head & Neck: Pupils 3 mm, non-reactive
Abdo: Distended	MSK/skin: Cool and dry
Other: PoCUS demonstrates some cardiac contractility, no palpable pulse but there is a PoCUS pulse. No lung sliding bilaterally. No pericardial effusion. No AAA. FAST neg.	

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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>specify type and whether infant/child/adult</i>)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Airway equipment: ET tube, suction, bougie, syringe, laryngoscope, video laryngoscope, stylet Magill forceps Soft FB to insert into the airway Cricothyrotomy kit
C. Required Medications
Epinephrine, calcium, bicarb, dextrose Heliox (optional)
D. Moulage
Soft FB in the airway just beyond the cords (we used a piece of foam tied to a string so that it can be extracted from the mannequin) Blood/secretions in the airway
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
Difficult to bag with OP and BVM On attempt at intubation a FB (food bolus) is identified sitting partially in the vocal cords Attempt to remove the FB causes it to slip through the cords

Section 4: Confederates and Standardized Patients

Confederate and Standardized Patient Roles and Scripts	
EMS	Deliver the patient with CPR in progress and provide collateral information
Bedside RN	Assist with starting IV, drawing labs and giving medications and updating with changes in vitals and preventing patient harm. Helpful and skilled.
RT	Assist with airway troubleshooting. Helpful and skilled.



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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: PEA HR: 80 BP: -- RR: -- O ₂ SAT: 60 % T: 36. 2°C GCS: 3	Patient unconscious	<u>Expected Learner Actions</u> ☐ Continue CPR and BVM ☐ Monitors, IV, limited vitals, cap BG ☐ Identify initial rhythm ☐ Verbalize potential Hs and Ts ☐ Labs, PoCUS ☐ Measure ET CO ₂ (25mmHg)	<u>Modifiers</u> Attempt to continue BVM→HR begins to fall into 40s and O ₂ sats do not improve Can consider other medications based on Hs and Ts (epi, naloxone, bicarb, calcium, thrombolysis) but none are effective at achieving ROSC <u>Triggers</u> All other actions move to stage 2	
2. Identify and treat hypoxia Same as state 1	Patient unconscious Airway is full of secretions and some blood	<u>Expected Learner Actions</u> ☐ Attempt to improve O ₂ sats with 2 handed BVM, flush FiO ₂ , repositioning, and a NPA in addition to OPA ☐ Can attempt an LMA ☐ Move to intubation during which you will identify the FB with laryngoscopy ☐ Manipulation of the FB with suction, Magills ☐ Can try to “apple-core” the FB with ET tube ☐ Force the FB into the right mainstem if other methods ineffective at retrieving the FB	<u>Modifiers</u> If no attempt to improve hypoxia, patient deteriorates into asystole LMA is still ineffective If cricothyrotomy attempted - ineffective (FB is below the level of the cricothyroid membrane) <u>Triggers</u> All actions complete move to stage 3	



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3. Airway secured Rhythm: sinus tach HR: 120s BP: 60/40 RR: -- O2 sat: improves to 75% T: 36.2 GCS: 3	Patient unconscious and now easy to bag Pulse palpable	<u>Expected Learner Actions</u> ☐ Start vasopressors/inotropes ☐ Consider post intubation sedation and analgesia ☐ Consult ENT, respiratory, or thoracic surgery for bronchoscopy	<u>Modifiers</u> <u>Trigger</u> All actions complete move to stage 4	
4. Stabilization Rhythm: sinus tach HR: 120s BP: 100/60 RR: -- O2 sats: 100% T36.2 GCS: 3	Patient remains unconscious	<u>Expected Learner Actions</u> ☐ Consult ICU ☐ Post-ROSC care (ECG, labs, x-ray) ☐ Consider cooling as still neurologically unresponsive post cardiac arrest	End of case	



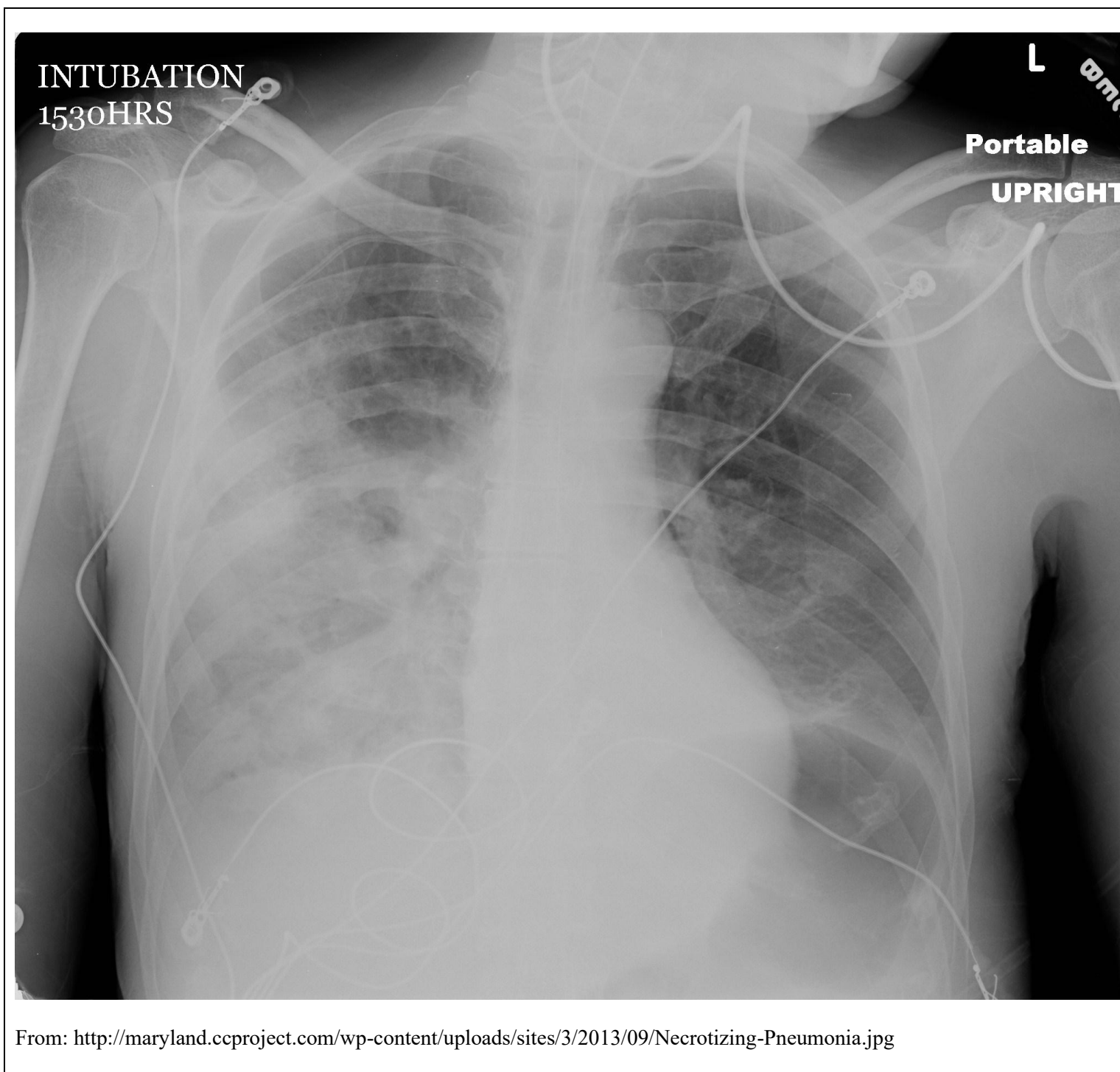
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Appendix A: Laboratory Results

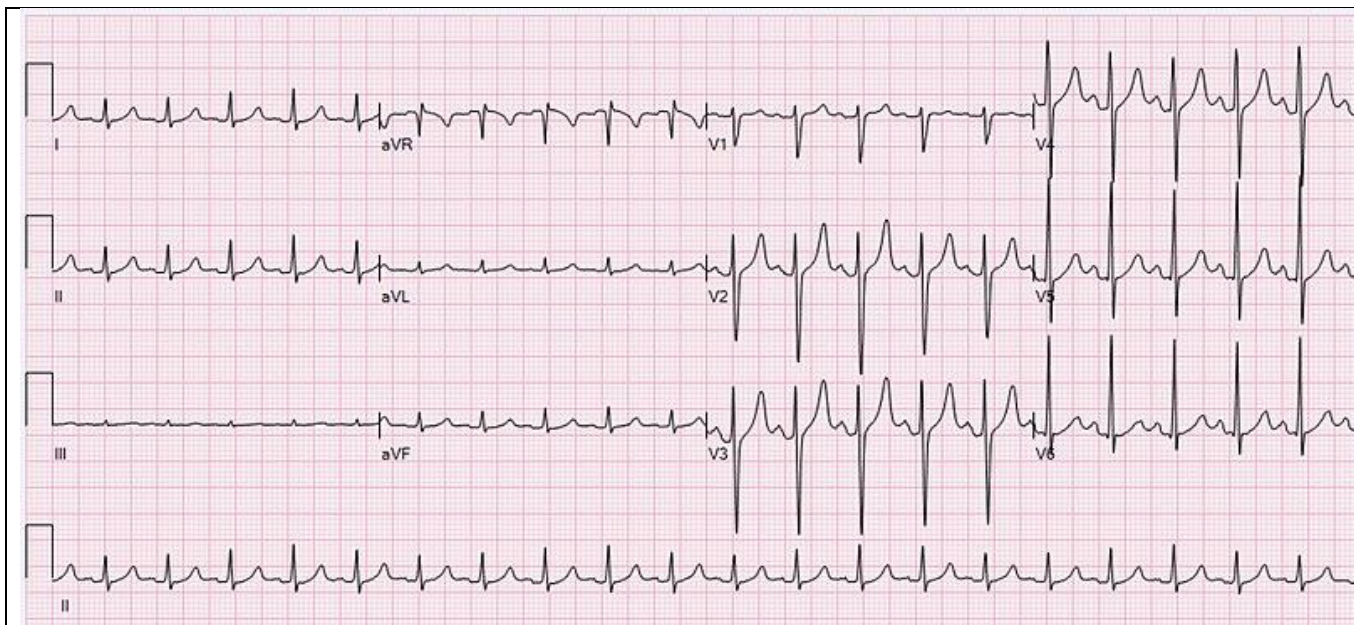
<u>Obtained POST ARREST if ROSC:</u> <u>CBC</u> WBC 20 Hgb 130 Plt 200 <u>Lytes</u> Na 149 K 6 Cl 110 HCO3 14 AG 24 Urea 8 Cr 150 Glucose 7 <u>VBG</u> pH 6.9 pCO2 60 pO2 60 HCO3 14 Lactate 9	<u>Cardiac/Coags</u> Trop 0.071 INR 1.1 <u>Biliary</u> AST 150 ALT 160 ALP 170 Bili 30 Lipase 200 <u>Tox</u> EtOH 80 ASA neg Tylenol neg Dig level neg Osmols 303
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures



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From Wikipedia (https://commons.wikimedia.org/wiki/File:ECG_Sinus_Tachycardia_125_bpm.jpg)

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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Key decision making points in the case:

1. Identify airway obstruction as the cause of cardiac arrest
2. Stepwise approach to securing the airway with a FB present
3. Utility of a cricothyrotomy with a FB present

Debriefing Topics:

1. Varying clinical presentation of airway obstruction from FB
2. Management of a partial vs. complete airway FB in an awake patient
3. Management of a partial vs. complete airway FB in an unconscious patient
4. Disposition of awake patients with a resolved FB
5. Consideration of antibiotics in aspiration pneumonia
6. Pediatric considerations

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